

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.6: Area of a Part of a Circle — Teacher Reference (Pre-filled)

SUMMARY TABLE: CORE MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.6: Area of a Part of a Circle
Driving Question	DRIVING QUESTION: How can we calculate the exact area of any part of a circle so that Kenyan farmers, engineers, and artisans can make accurate plans and avoid waste?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Area of an Annulus	An aerial photograph of a Laikipia County centre-pivot irrigation farm showing ring-shaped (annular), wedge-shaped, and overlapping watered zones that the full-circle formula $A = \pi r^2$ cannot fully explain.	The area of an annulus (the ring between two concentric circles of outer radius R and inner radius r) is found by subtracting the inner circle area from the outer circle area: $A = \pi R^2 - \pi r^2 = \pi(R^2 - r^2)$. This formula was derived using concentric circular card cut-outs and applied to an irrigation ring in Laikipia and a chapati-pan border.	Why $A = \pi r^2$ alone is insufficient for the farmer's planning needs, and how the annulus formula $A = \pi(R^2 - r^2)$ gives the exact area of the ring-shaped watered zone, enabling accurate calculation of planting area, fencing, and water budgets for the ring section of the farm.
Lesson 2 Area of a Sector: The Pivot's Sweep	Learners cut paper circles into sectors of 90° , 120° , 60° , and 45° , compared their areas as fractions of the whole circle, and completed a recording table showing how sector area changes with angle. They also examined an aerial-view sketch of the Laikipia centre-pivot field showing the 120° sweep of the pivot pipe.	A sector is the 'pie-slice' region of a circle bounded by two radii and an arc. Its area is a fraction of the full circle's area: $A = (\theta/360) \times \pi r^2$. The Laikipia pivot pipe sweeping 120° waters approximately 1675.52 m^2 of maize (one third of the full field). The Kisumu roundabout pedestrian refuge sector ($r = 6 \text{ m}$, $\theta = 75^\circ$) has an area of approximately 23.56 m^2 .	Because a full circle spans 360° , any sector with central angle θ covers exactly $\theta/360$ of the total area. Multiplying this fraction by the full-circle area (πr^2) gives the sector area formula $A = (\theta/360) \times \pi r^2$. This allows the Laikipia farmer to calculate precisely how much of the maize field has been watered at any point in the pivot's rotation, enabling accurate water budgeting and planting decisions.
Lesson 3 Arc Length and Sector Perimeter: Fencing the School Garden in Nakuru	Learners wrapped string around cardboard circle arcs of three different central angles (90° , 180° , 60°) and measured each arc length, recording results in a table that compared Arc÷Circumference with $\theta \div 360$, discovering the proportional relationship	The arc length of a sector is a fraction of the full circumference equal to the ratio of the central angle to 360° , giving the formula $l = (\theta/360) \times 2\pi r$; the full perimeter of a sector is this arc length plus the two radii, $P = l + 2r$.	Learners applied the arc length and sector perimeter formulae to three graded Kenyan contexts — an irrigation pivot in Laikipia, a curved market stall in Kisumu, and a sector-shaped maize plot — and updated their cumulative circle-decomposition diagrams

	directly from physical measurement.		to show arc length as a labelled curved boundary, connecting the mathematics directly to the fencing and planning needs of the anchoring phenomenon farmer.
Lesson 4 Area of an Annular Sector: Combining the Ring and the Wedge	Learners physically removed a small paper sector from a large paper sector of the same angle, producing a leftover annular-sector shape. They counted grid squares to estimate its area, then calculated $(\theta/360) \times \pi R^2$ and $(\theta/360) \times \pi r^2$ separately and subtracted — finding that the formula and the grid count agreed closely. They also examined measurements of a curved road-marking strip at Kisumu's Mega roundabout.	The area of an annular sector — the wedge-shaped ring swept by the outer irrigation pipe in Laikipia — is found by subtracting the area of the inner sector from the area of the outer sector: $A = (\theta/360) \times \pi(R^2 - r^2)$. This formula combines the annulus formula from Lesson 1 and the sector formula from Lesson 2. Importantly, each radius must be squared BEFORE subtracting — not the other way around.	Using the annular sector formula $A = (\theta/360) \times \pi(R^2 - r^2)$, the Laikipia farmer can calculate exactly how much of the outer irrigation ring has been watered at any angle, enabling accurate water budgeting, seed planning, and fencing estimates. The same formula solves the Kisumu road-marking problem, showing that one mathematical tool applies across agriculture and urban infrastructure.
Lesson 5 Area of a Segment of a Circle	Learners physically cut a paper sector ($r = 7$ cm, $\theta = 80^\circ$) along its chord, separated it into a triangle and a curved segment, estimated each area by counting squares on grid paper, then computed each area using formulas and compared the two methods.	A segment is the region between a chord and its arc. Its area equals the sector area minus the triangle area: $A = (\theta/360)\pi r^2 - \frac{1}{2}r^2 \sin \theta$. This was confirmed when the formula-derived value matched the grid-count estimate to within measurement error.	Learners applied the segment formula to find the rubber-seal area on a Turkana borehole hatch cover ($r = 35$ cm, $\theta = 110^\circ$) and the curved crack region on a Kamukunji sufuria lid ($d = 28$ cm, $\theta = 72^\circ$), connecting the formula back to the Laikipia farmer's challenge of calculating the exact area of a chord-cut region on the pivot field.
Lesson 6 Area of a Common Region Between Two Intersecting Circles	Learners overlapped two circular card cut-outs (Kericho tea-tray stencils) on an A3 sheet, traced the lens-shaped common region, identified the two intersection points P and Q, drew chord PQ, and measured the radii and centre-to-centre distance. They recorded scaled and real-world values in a results table and identified the lens as being composed of two circular segments.	The common region between two intersecting circles is a lens shape equal to the sum of two circular segments — one from each circle. The central angle for each segment is found using the cosine rule applied to the triangle formed by the two centres and one intersection point: $\cos \alpha = (r_1^2 + d^2 - r_2^2)/(2r_1d)$ and $\cos \beta = (r_2^2 + d^2 - r_1^2)/(2r_2d)$. Each segment area is calculated as $A = (2\theta/360)\pi r^2 - \frac{1}{2}r^2 \sin(2\theta)$, and the total lens area is the sum of both segment areas.	For the Laikipia farmer's two overlapping irrigation plots ($r_1 = 42$ m, $r_2 = 35$ m, $d = 49$ m), the central angle at A is approximately 88.8° and at B approximately 114.2° . The segment from Circle A has area ≈ 485.8 m ² , the segment from Circle B ≈ 662.3 m ² , giving a total shared (lens) area of approximately 1 148 m ² . This tells both farmers exactly how much land is jointly watered, enabling a fair split of water costs and preventing double-counting in planting and fencing plans.
Lesson 7 Mixed Application and Problem Solving: Combining Circle-Part Formulae	Learners examined four composite circle-part diagrams set in Rift Valley, Nairobi, Mombasa, and Laikipia contexts — each requiring the identification of two or more named sub-regions (annular sector, segment, common region, full sector) within a single combined figure.	Learners can decompose a composite circle-part figure into its constituent sub-regions, select and apply the correct formula for each region, combine the sub-areas using addition or subtraction, and express the solution as a structured multi-step calculation — skills directly needed for the Laikipia farmer's complete field-planning challenge.	Learners articulated why each sub-region formula is selected (based on shape structure, not memorisation), peer-assessed solutions using KICD rubric language, and connected every composite shape to a specific real-life scenario in Kenya, demonstrating that the driving question — how to calculate the exact area of any part of a circle — can now be

			answered for complex, combined shapes.
Lesson 8 Assessment, Model Revision and Consolidation — Answering the Farmer's Question	Learners observed their own completed cumulative diagrammatic models of the Laikipia farm displayed side by side — showing the progression from the single annulus of Lesson 1 to a fully decomposed farm plan annotated with all five circle-part formulae. During the gallery walk, they also observed which annotations were rated clearest by peers (green dots) and which were flagged as unclear (orange dots), providing immediate visual evidence of which formula representations communicate most effectively.	Learners consolidated all five formulae for the area of parts of a circle — annulus $A = \pi(R^2 - r^2)$, sector $A = (\theta/360)\pi r^2$, annular sector $A = (\theta/360)\pi(R^2 - r^2)$, segment $A = (\theta/360)\pi r^2 - \frac{1}{2}r^2 \sin \theta$, and the common region as the sum of two segment areas derived via the cosine rule — and demonstrated the ability to select and apply the correct formula for each physical sub-region of a composite figure, expressing results in the correct units with structured working.	The Laikipia farmer's planning problem — how to calculate the exact area of the ring between two irrigation pipes, the swept wedge at any point in the pivot's rotation, the annular band swept by the outer ring, the unsown region beside a drainage ditch, and the shared overlap with a neighbouring plot — is fully solved by decomposing the field into its constituent circle-part sub-regions and applying the appropriate derived formula to each. The driving question is answered: any part of a circle can be measured exactly, enabling Kenyan farmers, engineers, and artisans to plan resources precisely and avoid waste.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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